

EEM 413

PROJECT PROGRESS REPORT

FALL I

Project Title:	High-Speed Industrial "Black Box" Design for Vibration Analysis and Fault Diagnosis of Asynchronous Motors using STM32 Architecture
Thematic Area:	Embedded Systems Digital Signal Processing (DSP) Predictive Maintenance (PdM) Data Acquisition (DAQ)
Applying Institution:	Eskisehir Technical University - Department of Electrical and Electronics Engineering
Project Team:	Eren Özşen
Supporting Industry Partner:	-
Advisors:	Emin GERMEN

1. Work Package Activities

Work Package Description Form	
Work Package No:	1
Work Package Name:	Foundations and AFE Mastery
Starting Date:	16 th of February
Ending Date:	7 th of April
Planned Working Time (Days):	55 Days
Completed Working Time (Days):	50 Days
Deviation in Working Time (Days):	0 Days
Completed Working Time in This Term (Days):	50 Days
Work Package Manager:	Eren Özşen
What is the status of this work package at the beginning of this term? At the beginning of this term, the work package was entirely in the theoretical research and component acquisition phase. The core hardware components, including the STM32F407G-DISC1 development board, TL084 op-amps, and piezoelectric transducers, were available. However, the custom Analog Front-End (AFE) circuit design and the configuration of STM32 hardware registers (ADC, DMA, Clock Tree) had not yet been initiated.	
Which work package activities have you worked on during this term? Who have contributed each of these activities? Explain how she/he has contributed as well. During this term, all activities for Work Package 1 were exclusively conducted by Eren Özşen. The primary objective was to build a robust foundation in STM32 hardware architecture and peripheral integration. The work package commenced with extensive theoretical research. I studied the STM32F407G-DISC1 datasheets and reference manuals to understand the ARM Cortex-M4 architecture, clock tree mechanics, and peripheral functionalities. Simultaneously, I familiarized myself with the STM32CubeMX configuration tool and the STM32CubeIDE development environment. To transition from theory to practice, I developed initial firmware projects. In my first major test project, I combined Hardware Timers and External Interrupts (EXTI). I designed a sequential 'walking LED' algorithm where the shifting speed is dynamically controlled via the onboard user button by modifying the Timer's Auto-Reload Register (ARR) values on the fly within the EXTI callback. To further deepen my understanding of the board's capabilities and external sensor communication, I focused on interfacing with the onboard MEMS accelerometer for my second project. I researched and integrated open-source hardware abstraction libraries (specifically for LIS302DL and LIS3DSH sensors) into my STM32CubeIDE project. I developed a firmware routine that first queries the board to dynamically detect the specific onboard MEMS chip model and initializes it. Subsequently, I implemented a tilt-detection algorithm that reads the X, Y, and Z-axis data. The code dynamically lights up the corresponding directional LEDs based on the board's physical orientation (tilt). Additionally, to monitor the raw sensor data in real-time, I redirected the printf function via the ITM (Instrumentation Trace Macrocell) to output the 3-axis readings directly to the	

debugging console. This step significantly enhanced my practical knowledge of external sensor integration and real-time data monitoring on the STM32 platform. As the final phase of Work Package 1, I initiated the hardware interfacing of the piezoelectric sensors by constructing the custom Analog Front-End (AFE). Anticipating the final 3-axis design, I selected the TL084 quad operational amplifier, utilizing a single op-amp channel for this initial test. The op-amp was powered by an external 9V battery. To enable the STM32's single-ended ADC to read the alternating AC voltage generated by the piezo sensor, I established a DC bias (virtual ground) by creating a voltage divider with two identical resistors connected to the board's 3.3V output. The sensor was conditioned using a parallel resistor and a capacitor (approx. 10nF and 100kΩ) to filter the high-impedance signal. On the firmware side, the main.c implementation was kept minimal, utilizing HAL_ADC_Start_DMA to transfer data into a 1024-element adc_buffer array. Using the IDE's live debug expressions, I monitored the buffer's first element. In a resting state, the ADC value remained stable around 1800-1900, successfully validating the 1.65V DC bias point. Upon lightly tapping the piezo sensor, the value instantly peaked at the 12-bit limit of 4095, confirming that the AFE correctly scaled the mechanical vibrations into digital values. This milestone marked the successful completion of the AFE prototyping.

Reasons for the deviation from the project plan, if any?

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Outputs:

- Verified high-impedance Analog Front-End (AFE) prototype circuit for a single piezoelectric sensor on a breadboard.
- Successful implementation of the 1.65V DC bias (virtual ground) and practical validation of the sensor's peak response (0-4095) via STM32 ADC live debugging.
- Working firmware for real-time 3-axis tilt detection using the onboard MEMS accelerometer, with data successfully monitored via the ITM console.
- Established solid practical foundation and working codebase for STM32 hardware peripherals (Timers, EXTI) and development tools (CubeMX, CubeIDE).

Work Package Description Form	
Work Package No:	2
Work Package Name:	High-Speed Acquisition Pipeline
Starting Date:	8 th of April
Ending Date:	12 th of May
Planned Working Time (Days):	45 Days
Completed Working Time (Days):	35 Days
Deviation in Working Time (Days):	0 Days
Completed Working Time in This Term (Days):	35 Days
Work Package Manager:	Eren Özşen
What is the status of this work package at the beginning of this term? At the beginning of this term, the high-speed data acquisition pipeline was entirely in the conceptual planning phase. The required external hardware components for data storage, specifically the SD card adapter module, had not yet been procured. Furthermore, the firmware architecture for synchronized multi-channel ADC sampling and high-throughput data writing protocols (SDIO/SPI) was strictly theoretical, with no codebase or physical connections established.	
Which work package activities have you worked on during this term? Who have contributed each of these activities? Explain how she/he has contributed as well. During this term, the activities for Work Package 2 were conducted by Eren Özşen. For the data acquisition phase, I advanced to reading raw vibration data from a single piezoelectric sensor. I connected the sensor's AFE output to pin PA0, configured as ADC1 IN0. To achieve a precise 10 kHz sampling rate without burdening the CPU, I configured Timer 3 (TIM3) with a prescaler of 83 and a counter period of 99, utilizing the 'Update Event' as the Trigger Output (TRGO). In the ADC settings, I enabled 'Scan Conversion' and, most importantly, 'DMA Continuous Requests'. This hardware-level synchronization ensured that the ADC sampled the sensor exactly at 10 kHz and automatically transferred the converted data directly to the memory buffer via DMA, completely bypassing CPU intervention. To ensure continuous data flow without manual re-triggering, the ADC DMA request was explicitly configured to 'Circular' mode. Following the successful single-channel validation, the system was scaled to a full 3-axis acquisition architecture. In STM32CubeMX, the ADC 'Number Of Conversions' was increased to 3. A critical hardware optimization was implemented by increasing the 'Sampling Time' for each ADC channel to 144 cycles; this effectively eliminated the analog crosstalk (ghosting) between the multiplexed X, Y, and Z axes, ensuring clean and isolated data acquisition for all three sensors.	
Reasons for the deviation from the project plan, if any? There is a planned deviation regarding the data logging protocol. The original project proposal specified the use of the SDIO (4-bit mode) interface for high-throughput data writing. However, due to severe procurement delays of the SDIO reader module from overseas suppliers, waiting for the component posed a major risk to the project timeline. After consulting with my advisor, a contingency plan was activated. The data acquisition pipeline was successfully restructured to utilize the	

standard SPI protocol combined with the FatFs file system. This deviation did not compromise the project goals; instead, it demonstrated agile problem-solving to establish a reliable 'Black Box' logging infrastructure within the given timeframe.

Outputs:

- Synchronized and jitter-free 3-channel ADC acquisition pipeline, accurately triggered by Timer 3 (TIM3) at a rate of 10 kHz per channel.
- Successful implementation of the DMA engine in 'Circular' mode, enabling continuous high-throughput data transfer directly to memory without CPU intervention.
- Complete elimination of analog crosstalk (ghosting) between the multiplexed X, Y, and Z piezoelectric channels through hardware sampling time optimization (144 cycles).

Work Package Description Form	
Work Package No:	3
Work Package Name:	Firmware Verification & Calibration
Starting Date:	8 th of April
Ending Date:	12 th of May
Planned Working Time (Days):	45 Days
Completed Working Time (Days):	35 Days
Deviation in Working Time (Days):	0 Days
Completed Working Time in This Term (Days):	35 Days
Work Package Manager:	Eren Özşen
What is the status of this work package at the beginning of this term? At the beginning of this term, the firmware verification and calibration package was strictly in the conceptual design phase. The complex software architecture required for the DMA-driven Ping-Pong (Double) Buffering logic had not yet been developed. Furthermore, the standalone "Black Box" logging infrastructure, including the integration of the FatFs file system for gapless data storage to an SD card, existed only as a theoretical framework with no initial codebase or calibration protocols established.	
Which work package activities have you worked on during this term? Who have contributed each of these activities? Explain how she/he has contributed as well. During this term, the activities for Work Package 3 were conducted by Eren Özşen. The primary focus was establishing the standalone 'Black Box' logging infrastructure. The initial step involved hardware procurement. While a standard MicroSD card and an SD card adapter were readily available, the SDIO card reader module had to be ordered from abroad. Due to significant shipping delays, I consulted with my advisor and made a strategic decision to pivot to a locally available, standard SPI-based MicroSD card reader to keep the project on schedule. For the first practical integration, I interfaced the SPI card reader directly with the STM32F407G-DISC1. To prevent any SPI bus conflicts, I implemented a critical hardware-level precaution in the firmware: I pulled the PE3 pin HIGH to actively disable (chip-deselect) the onboard MEMS accelerometer, which shares the same SPI lines. Following this, I integrated the FatFs file system library. I developed a preliminary test firmware to verify the SPI communication and file system operations. The code successfully mounted the SD card, created a text file named 'deneme.txt', and wrote the string 'STM32 Industrial Black Box - SD Kart Testi Basarili!' into it. The entire process (Mount, Open, Write, and Error states) was visually debugged using the onboard LEDs. This successful test validated the core storage infrastructure, paving the way for high-speed raw piezoelectric data logging. Building upon the initial text file test, I developed the continuous high-speed data logging architecture for the raw sensor data. I allocated a 4096-element array (adc_buffer) and implemented a DMA-driven Ping-Pong (Double) Buffering strategy. Utilizing the HAL_ADC_ConvHalfCpltCallback and HAL_ADC_ConvCpltCallback functions, the firmware tracks the buffer state. In the main loop, while the DMA fills one half of the buffer in the background, the CPU	

utilizes FatFs (f_write and f_sync) to safely write the other half to a continuous binary file named 'PIEZO.BIN'. During this integration, I resolved a critical SPI bus conflict: to keep the wiring organized, I chose PA4 as the Chip Select (CS) pin due to its proximity to the SPI pins (PA5, PA6, PA7). However, this shared the bus with the onboard MEMS accelerometer. Pulling the PE3 pin HIGH effectively disabled the onboard sensor's CS, ensuring uninterrupted SD card communication. The entire system's operational status—including successful gapless writing and potential SPI errors—was monitored in real-time using the onboard diagnostic LEDs.

To accommodate the tripled data throughput generated by the new 3-axis acquisition setup, the logging infrastructure required calibration. The SPI1 baud rate was significantly increased by lowering its prescaler from 32 to 8. Furthermore, the firmware's memory management was updated to handle a larger 6144-element array (BUFFER_SIZE). Utilizing the established DMA-driven Ping-Pong logic, the system successfully and continuously wrote the high-speed interleaved 3-channel data stream into a new binary file named 'PIEZO3.BIN', finalizing the fully operational Black Box recording capability.

Reasons for the deviation from the project plan, if any?

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Outputs:

- Fully functional standalone "Black Box" logging infrastructure utilizing the standard SPI protocol and the FatFs file system.
- Robust DMA-driven Ping-Pong (Double) Buffering architecture successfully managing a 6144-element data array without buffer overruns.
- Gapless and continuous recording of high-speed interleaved 3-channel vibration data into a raw binary format (PIEZO3.BIN) directly to the SD card.
- Development of custom Python scripts (tekoku.py and ucluoku.py) that successfully de-interleave the binary data and visualize the 3-axis waveforms, providing final verification of absolute signal integrity.

2. Status of Intermediate Outputs

Intermediate Outputs			
Output	Expected Date	Completion Date	Reason for any deviation?
Verified High-Impedance AFE Prototype	Mid-April	7th of April	None. The theoretical research and prototyping were completed ahead of schedule.
Gapless 3-Channel Vibration Log File	Mid-May	12th of May	SDIO module procurement delay. The protocol was successfully switched to SPI with FatFs to ensure on-time completion.
Complete Infrastructure Documentation (Black Box)	End of May	12th of May	None. The infrastructure was completed alongside the data logging phase successfully.

3. Changes in Project Plan

Do you have any changes in the goals and activities of your project? Explain.

There are no changes in the fundamental goals of the project. The primary objective remains to design a high-speed 'Industrial Black Box' for multi-axis vibration analysis using the STM32 architecture. However, a minor technical activity change occurred regarding the SD card interfacing method. Due to component procurement issues, the planned activity of writing the SDIO protocol driver was replaced with the integration of the SPI protocol and the FatFs file system.

Do you have any changes in your project plan?

Yes, a strategic change was implemented in the project plan to mitigate a critical supply chain risk. The planned reliance on an imported SDIO card reader module was replaced with a locally sourced, standard SPI-based MicroSD adapter. This proactive change in the technical plan prevented potential weeks of delay. Consequently, it allowed the project to strictly adhere to the original timeline for Work Packages 2 and 3, ensuring the successful and timely completion of the gapless data logging infrastructure during the current term.

4. Project Management

Project Overview

During this term, I have successfully executed Work Packages 1, 2, and 3 of the 'High-Speed Industrial Black Box' project. The primary goal of this phase was to construct a reliable, high-bandwidth data acquisition (DAQ) infrastructure. By integrating 3-axis piezoelectric sensors, a custom Analog Front-End (AFE), and the STM32 microcontroller, a standalone system capable of continuously recording high-frequency vibration data was established. This accomplishment successfully lays the essential hardware and firmware foundation for the advanced predictive maintenance and fault diagnosis analysis planned for the upcoming term.

Work Packages

Work Package 1: In this initial package, the focus was on understanding the STM32F407G-DISC1 hardware architecture and establishing the analog sensor interface. Theoretical research was followed by practical firmware test projects, including timer-based interrupts and onboard MEMS sensor communication. The package culminated in the successful breadboard prototyping of a custom Analog Front-End (AFE) circuit using TL084 operational amplifiers, which properly conditioned the high-impedance piezoelectric signals for the STM32 ADC.

Work Package 2: The core of this work package was developing a synchronized, jitter-free data acquisition system. The firmware was configured to sample three independent piezoelectric channels simultaneously at 10 kHz using Timer 3 triggers and the DMA 'Circular' mode. A critical engineering challenge—analogue crosstalk between ADC channels—was successfully resolved by optimizing the hardware sampling cycles, ensuring clean, isolated data acquisition across the X, Y, and Z axes without burdening the CPU.

Work Package 3: This package focused on establishing the gapless data logging infrastructure. To mitigate hardware supply delays, the storage protocol was agilely pivoted from SDIO to a standard SPI and FatFs integration. A DMA-driven Ping-Pong (Double) Buffering architecture was developed to handle the massive data throughput seamlessly. The system successfully managed to write the interleaved 3-channel vibration data continuously into a raw binary file ('PIEZO3.BIN'), which was later verified for absolute signal integrity using custom Python visualization scripts.

Technical Challenges and Solutions

During the development of the High-Speed Industrial Black Box, several hardware and firmware challenges were encountered and systematically resolved:

- **Procurement Delays:** The original plan required an SDIO card reader module, which faced severe shipping delays. **Solution:** To maintain the project timeline, a strategic pivot was made to a locally available standard

SPI-based MicroSD adapter, which was successfully integrated using the FatFs file system.

- **Hardware SPI Bus Conflict:** The chosen SPI lines for the SD card adapter were shared with the STM32's onboard MEMS accelerometer, causing communication interference. **Solution:** A hardware-level precaution was implemented in the firmware by pulling the PE3 pin HIGH. This actively disabled (chip-deselected) the onboard sensor, ensuring uninterrupted and stable SD card logging.
- **Analog Signal Crosstalk (Ghosting):** Upon scaling to a 3-axis acquisition architecture, the ADC readings exhibited crosstalk, where the Z-axis sensor mimicked the Y-axis due to the rapid switching of the ADC multiplexer. **Solution:** A critical hardware optimization was applied by increasing the ADC sampling time to 144 cycles. This allowed the internal hold capacitor sufficient time to stabilize during channel transitions, completely isolating the analog signals.
- **High-Speed Data Bottlenecks:** Sampling 3 channels continuously at 10 kHz risked CPU overload and buffer overruns. **Solution:** A DMA-driven Ping-Pong (Double) Buffering architecture was developed to manage a 6144-element array. Additionally, the SPI baud rate was significantly increased (prescaler reduced from 32 to 8), enabling gapless binary data writing to the SD card without CPU intervention.

Project Progress and Milestones

Substantial progress has been achieved in Work Packages 1, 2, and 3, successfully establishing the project's core hardware and firmware foundation. A proactive shift to an SPI-based logging architecture prevented potential supply chain delays, allowing the project to stay strictly on schedule.

The most significant milestone reached this term is the fully operational "Industrial Black Box" prototype. The system successfully acquires 3-axis high-frequency piezoelectric data using DMA Ping-Pong buffering and writes it gaplessly to an SD card. Furthermore, early progress was made by developing Python scripts to parse and visualize the recorded binary data, validating absolute signal integrity.

Moving forward, the focus will shift from laboratory prototyping to field data collection. The upcoming milestones include gathering extensive vibration signatures from healthy industrial motors and motors with seeded mechanical faults during the summer period. Following data collection, the project will advance to implementing STFT (Short-Time Fourier Transform) and generating Spectrograms to visualize dynamic frequency changes. This will lay the groundwork for developing the final diagnostic identification logic based on the extracted visual fingerprints.

Conclusion

In summary, the 'High-Speed Industrial Black Box' project has successfully completed its foundational phase. Despite initial supply chain setbacks regarding the SDIO module, a strategic pivot to an SPI-based logging architecture ensured the project remained strictly on schedule. Critical engineering challenges, including analog signal crosstalk and SPI bus conflicts, were successfully diagnosed and resolved through precise hardware-level firmware optimizations. The project has yielded a fully functional, 3-axis, high-speed data acquisition prototype capable of

gapless recording via DMA Ping-Pong buffering. Moving forward, the focus will shift towards extensive field data collection from industrial motors and advanced Spectrogram analysis. The robust infrastructure established this term provides a highly reliable and professional foundation for achieving the ultimate goal of developing a cost-effective predictive maintenance tool.

5. Learning Outcomes

During the development of the Industrial Black Box Project, I gained profound practical experience by seamlessly integrating analog hardware design with advanced embedded software. This process provided me with a deep, comprehensive understanding of the ARM Cortex-M4 architecture and the STM32 microcontroller ecosystem.

A significant portion of my learning centered around Analog Front-End (AFE) design. I practically learned how to handle high-impedance signals from piezoelectric sensors, design virtual ground (DC bias) circuits, and utilize operational amplifiers (TL084) to properly condition raw analog signals for accurate ADC digitization.

On the software front, I significantly advanced my embedded C programming skills. Moving beyond simple CPU-blocking delays, I mastered hardware-level synchronization using internal Timers and the Direct Memory Access (DMA) engine. Implementing a Ping-Pong (Double) Buffering architecture taught me how to manage massive data throughput efficiently without overloading the CPU.

Debugging complex hardware-software interactions was another crucial learning outcome. I developed strong problem-solving skills by identifying and resolving intricate issues such as ADC multiplexer crosstalk (ghosting) by manipulating sampling cycles, and managing SPI bus sharing conflicts between the onboard MEMS sensor and the SD card reader.

Furthermore, I bridged the gap between embedded systems and data science. By writing custom Python scripts utilizing NumPy and Matplotlib, I learned how to parse raw binary files generated by the microcontroller and visualize the multi-axis vibration data, effectively verifying the hardware's signal integrity.

Lastly, managing this project entirely on my own enhanced my project management and crisis response skills. Adapting to supply chain delays by rapidly pivoting from an SDIO protocol to an SPI and FatFs integration demonstrated the importance of agile problem-solving in real-world engineering projects.